

COMP 10280

Programming I (Conversion)

Practical Sheet 15

Dé hAoine, 9 Deireadh Fómhair 2026

For each of the following questions, write an algorithm in pseudocode first before writing a Python program. Submit your algorithms in pseudocode as well as your Python programs. When writing functions, use one-line or multi-line docstrings, as appropriate, to document your functions.

1. Consider the following series, defined recursively as follows:

$$f(n) = \begin{cases} 2 & n = 1 \\ n + f(n - 1) & n > 1 \end{cases}$$

- (a) Write a recursive function that takes as its single argument an integer ≥ 1 and returns as a result that term from the above series.
- (b) Write a program that prompts the user for a series of integers. For each number entered the program should check that it is greater than or equal to 1. If it is, it calls the function defined in part (a) and prints out the result. The program should stop when a zero or a negative number is entered.
- (c) In your function, include some print statements that allow you to see the operation of the recursion and its progress towards the base case.

Save this program as p15p1.py.

2. Write a program that prompts the user for a positive integer (≥ 1) and, using the function developed in Question 1(a), prints out that number of terms of the series defined in Question 1. (If you had included print statements when developing your function (Question 1(c)), you can discard them for this question.)

Save this program as p15p2.py.

3. Consider the following series, defined recursively as follows:

$$f(n) = \begin{cases} 1 & n = 1 \\ f(n - 1) + 2^{n-1} & n > 1 \end{cases}$$

- (a) Write a recursive function that takes as its single argument an integer ≥ 1 and returns as a result that term from the above series.
- (b) Write a program that prompts the user for a series of integers. For each number entered the program should check that it is greater than or equal to 1. If it is, it calls the function defined in part (a) and prints out the result. The program should stop when a zero or a negative number is entered.
- (c) In your function, include some print statements that allow you to see the operation of the recursion and its progress towards the base case.

Save this program as p15p3.py.

4. Write a program that prompts the user for a positive integer (≥ 1) and, using the function developed in Question 3(a), prints out that number of terms of the series defined in Question 3. (If you had included print statements when developing your function (Question 3(c)), you can discard them for this question.)

Save this program as p15p4.py.

5. Consider the following series, defined recursively as follows:

$$f(n) = \begin{cases} 13 & n = 0 \\ 8 & n = 1 \\ f(n-2) + 13 \times f(n-1) & n > 1 \end{cases}$$

- (a) Write a recursive function that takes as its single argument an integer ≥ 0 and returns as a result that term from the above series (Assume that the 0th term is the first term).
- (b) Write a program that prompts the user for a series of integers. For each number entered the program should check that it is greater than or equal to 0. If it is, it calls the function defined in part (a) and prints out the result. The program should stop when a negative number is entered.
- (c) In your function, include some print statements that allow you to see the operation of the recursion and its progress towards the base cases.

Save this program as p15p5.py.

6. Write a program that prompts the user for a non-negative integer (≥ 0) and, using the function developed in Question 5(a), prints out that number of terms of the series defined in Question 5. (If you had included print statements when developing your function (Question 5(c)), you can discard them for this question.)

Save this program as p15p6.py.

**Please upload your work to
the Brightspace site before Sunday
evening.**

**You should keep a copy of your programs
for your portfolio.**